

Definition

Let (Y, X) be a random vector. The conditional expectation of Y given $X=x$ is called Regression of Y on X .

$$\hat{y} = g(x, \beta) = E(Y|X=x)$$

y : response variable (Stochastic / Random Variable)
 x : regressor variable (Stochastic / Non-stochastic)

(#)

A regression is called linear Regression if the relation b/w y & x is linear in parameter β .

→ $Y = X\beta + \epsilon$ generalized structure for all linear models

→ Generalized Linear Model:

$$GLM = g(X\beta) + \epsilon$$

For linear models: $g(\cdot)$ is an identity func'n $\Rightarrow g(X\beta) = X\beta$

LINEAR
$y = \beta_0 + \beta_1 x + \epsilon$
$y = \beta_0 + \beta_1 x + \epsilon$

NON-LINEAR
$y = \frac{1}{\beta_0 + \beta_1 x} + \epsilon$
$y = \beta_0 \cos(\beta_1 + \beta_2 x) + \epsilon$

Definition

Considering least squared error $S(\beta_0, \beta_1) = \sum_i (y_i - \beta_0 - \beta_1 x_i)^2$

solving $\frac{\partial S}{\partial \beta_0} = 0$, $\frac{\partial S}{\partial \beta_1} = 0$

$$\hat{\beta}_1 = \frac{\sum x_i y_i - \frac{(\sum y_i)(\sum x_i)}{n}}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}} = \frac{S_{xy}}{S_{xx}}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$S_{xy} = \sum (y_i \bar{y})(x_i - \bar{x})$
$S_{xx} = \sum (x_i - \bar{x})^2$
$\bar{y} = \sum y_i / n$
$\bar{x} = \sum x_i / n$

PREDICTED LINE $\Rightarrow \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$

PROOF: $S(\beta_0, \beta_1) = \sum (y_i - \beta_0 - \beta_1 x_i)^2$

→ $\frac{\partial S}{\partial \beta_0} = 0 \Rightarrow \sum -2(y_i - \beta_0 - \beta_1 x_i) = 0$

$$\sum y_i - n\beta_0 - \beta_1 \sum x_i = 0 \quad \text{--- (I)}$$

→ $\frac{\partial S}{\partial \beta_1} = 0 \Rightarrow \sum -2(y_i - \beta_0 - \beta_1 x_i)(x_i) = 0$

$$\Rightarrow \sum y_i x_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 = 0 \quad \text{--- (II)}$$

From (I) $\beta_0 = \frac{\sum y_i - \beta_1 \sum x_i}{n}$

$$\Rightarrow \sum y_i x_i - \left(\frac{\sum y_i - \beta_1 \sum x_i}{n} \right) \sum x_i - \beta_1 \sum x_i^2 = 0$$

$$\Rightarrow \sum y_i x_i - \frac{\sum y_i \sum x_i}{n} + \frac{\beta_1 (\sum x_i)^2}{n} - \beta_1 \sum x_i^2 = 0$$

$$\Rightarrow \frac{\left(\sum y_i x_i - \frac{\sum y_i \sum x_i}{n} \right)}{\sum x_i^2 - \frac{(\sum x_i)^2}{n}} = \hat{\beta}_1$$

$$\Rightarrow \frac{\left(\sum y_i x_i - n \bar{y} \bar{x} \right)}{\sum x_i^2 - n \bar{x}^2} = \hat{\beta}_1$$

$$\Rightarrow \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = \hat{\beta}_1 = \frac{s_{xy}}{s_{xx}}$$

$$\hat{\beta}_0 = \frac{\sum y_i - \beta_1 \sum x_i}{n} = \bar{y} - \hat{\beta}_1 \bar{x}$$

Residual

$$\text{residual} \leftarrow \epsilon_i = y_i - \hat{y}_i$$

$\downarrow$ observed value $\downarrow$ fitted value

Matrix Representation

$$\text{data} \downarrow D = \{(x_i, y_i) \mid x_i \in \mathbb{R}, y_i \in \mathbb{R}\}$$

x : non-stochastic
 y : stochastic

$$(*) \quad \underset{\sim}{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = (y_1, y_2, \dots, y_n)^T$$

$$(*) \quad \underset{\sim}{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}$$

$$(*) \quad X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix} = \begin{bmatrix} \underset{\sim}{1} & \underset{\sim}{x} \end{bmatrix} \rightarrow \text{This depicts that we have 1 feature & n samples.}$$

$$(*) \quad \underset{\sim}{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix} = (\epsilon_1, \epsilon_2, \dots, \epsilon_n)^T$$

(SIMPLE LINEAR REGRESSION)

$$\text{Regression Equation: } \underset{\sim}{y} = X \underset{\sim}{\beta} + \underset{\sim}{\epsilon}$$

Say if we have 2 features x_1, x_2 our X matrix will be modified:-

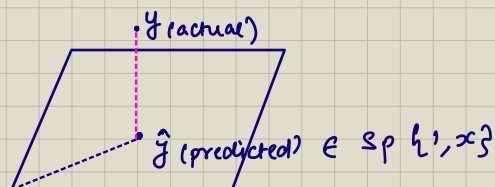
$$X = \begin{bmatrix} 1 & x_1 & x_2 \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{21} \\ 1 & x_{12} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{1n} & x_{2n} \end{bmatrix} \quad \text{MULTIPLE LINEAR REGRESSION}$$

Modifying some features x_1, x_2, x_1^2, x_2^2 we get

$$X = \begin{bmatrix} 1 & x_1 & x_2 & x_1 x_2 & x_1^2 & x_2^2 \end{bmatrix} \quad \text{POLYNOMIAL REGRESSION}$$

Terms from Linear Algebra:

- ① Vector space
- ② Sub-space
- ③ Basis
- ④ Span
- ⑤ Linear Independence



(*) It often happens that there are features which can be expressed as a linear combination of other features. Existence of such repeated vectors may lead to $|\mathbf{D}| = 0$. (MULTIPLE LINEARITY PROBLEM)

(*) Two vectors are said to be orthogonal if $u^T v = 0$ ($u, v \in V$)

(*) If $S \subseteq V$ is a subspace then the orthogonal complement of S , denoted by $S^\perp$ is defined as:-

$$S^\perp = \{ v \mid v \in V, u^T v = 0, \forall u \in S \}$$

$$\dim(S^\perp) = \dim(V) - \dim(S), \quad S \cap S^\perp = \{0, 0\}$$

Properties:-

- Basis is not unique.
- Elements of a basis need not be orthogonal to each other.
- Linear Independence does not imply orthogonality.
- Orthogonality implies independence (OII!!)
- Orthogonal vectors with unit length are called orthonormal vectors.

Definition:

If $S \subseteq V$, then the projection matrix of subspace S is P_S satisfying:-

$$\begin{aligned} \rightarrow P_S v &= v, \quad v \in S \\ \rightarrow P_S v &\in S, \quad \forall v \in V \end{aligned}$$

(*) A projection matrix P_S is an orthogonal projection matrix of subspace $S \subseteq V$ if $(I - P_S)$ is a projection matrix of $S^\perp \subseteq V$ too.

(*) If $\{v_1, v_2, \dots, v_k\}$ is an orthonormal basis of the subspace $S \subseteq V$ then the orthogonal projection matrix of S is

$$P_S = \sum v_i v_i^T$$

In regression eventually $\hat{y}$ becomes the orthogonal projection of $y \in \mathbb{R}^n$ in the sub-space $S = \text{Sp}[1, x]$

Prediction: $\hat{y} \in \text{Sp}[1, x]$ i.e. $\hat{y} = P_x y$

error: $\tilde{y} - \hat{y} = (I - P_x) y$

Definition:

$$A = [a_1, a_2, a_3, \dots, a_n]$$

$a_1, a_2, \dots, a_n \rightarrow$ columns of matrix A .

$$C(A) = \text{Sp} \{a_1, a_2, \dots, a_n\} = \{Ax \mid x \in \mathbb{R}^n\}$$

$$R(A) = C(A^T)$$

Properties:

- ① $C(A:B) = C(A) + C(B)$
- ② $C(AB) \subseteq C(A)$
- ③ $\dim[C(A)] = \text{Rank}(A)$
- ④ $C(AA^T) = C(A) \Leftrightarrow \text{Rank}(AA^T) = \text{Rank}(A)$
- ⑤ If A has n rows then $\dim(C(A)^\perp) = n - \text{Rank}(A)$

Definition:

A square symmetric matrix $A = (A_{ij})_{n \times n}$ is said to be

a) positive definite (p.d.)

$$x^T A x > 0 \quad \forall x \neq 0 \in \mathbb{R}^n$$

b) positive semi-definite (p.s.d.)

$$x^T A x \geq 0 \quad \forall x \neq 0 \in \mathbb{R}^n$$

If A is p.d. $|A| > 0$

" A is p.s.d. $|A| \geq 0$

Definition:

A matrix G is said to be a generalized inverse of a matrix A if $AGA = A$. G is denoted by A^- .

→ if A is $m \times n$, A^- is $n \times m$

→ A^- is not unique

→ For a matrix A the projection matrix of $C(A) = AA^-$

→ For a matrix A the orthogonal projection matrix of $C(A)$ is

$$A(A^T A)^- A^T$$

NOTE: In regression the prediction $\hat{y}$ and the error $y - \hat{y}$ are orthogonal to each other.

Definition:

Let $x = (x_1, x_2, \dots, x_n)^T$ is a random vector

$$E(\vec{x}) = (E(x_1), E(x_2), \dots, E(x_n))'$$

→ similarly if $Y = ((y_{ij}))_{m \times n}$ is a random matrix

$$E(Y) = ((E(y_{ij})))_{m \times n}$$

(#) By linearity properties of expectation it can be stated that

$$E(A\vec{x}) = AE(\vec{x}) \quad E(\vec{x}B) = (E\vec{x})B$$

A : $k \times n$ matrix

$\vec{x}$: random vector

B : $1 \times j$ matrix

(*) An important rule: $E(x)$ should be finite and exist for each component.

Definition:

Dispersion Matrix also called Variance - Covariance Matrix

$$D(\vec{x}) = ((\text{Cov}(x_i, x_j)))_{n \times n} = \text{Cov}(\vec{x}, \vec{x}) = E[(\vec{x} - E(\vec{x}))(\vec{x} - E(\vec{x}))^T]$$

$\vec{x}$: random vector

For a random variable X, Y

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E[X Y - X E(Y) - Y E(X) + E(X) E(Y)]$$

$$= E(XY) - E(X)E(Y) - E(Y)E(X) + E(X)E(Y)$$

$$= E(XY) - E(X)E(Y)$$

$$\text{Cov}(X, X) = E(X^2) - [E(X)]^2$$

$$= \text{Var}(X)$$

Now, we have $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$ a random vector

$$D(\vec{x}) = E[(\vec{x} - E(\vec{x}))(\vec{x} - E(\vec{x}))^T] = E[(\vec{x} - \vec{\mu})(\vec{x} - \vec{\mu})^T]$$

$$= E(\vec{x}\vec{x}^T) - E(\vec{x})[E(\vec{x})]^T = E(\vec{x}\vec{x}^T) - \vec{\mu}\vec{\mu}^T$$

$$\text{Cov}(\vec{X}) = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_p) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \text{Cov}(X_p, X_2) & \cdots & \text{Var}(X_p) \end{bmatrix}$$

$$\begin{aligned} \rightarrow \text{Cov}(U_p, V_q) &= ((\text{Cov}(U_i, V_j)))_{p \times q} \\ \rightarrow E(x+b) &= E(x) + b \quad \text{location equivariant} \\ \rightarrow D(x+b) &= D(x) \quad \text{location invariant} \\ \rightarrow \text{Cov}(x+b, y+c) &= \text{Cov}(x, y) \end{aligned}$$

Definition:

Let $\mathbf{X}$ be a random vector with n -components such that $E(\mathbf{X}) = \mu$ and $D(\mathbf{X}) = \Sigma$ then

1 $E(l^T \mathbf{X}) = l^T \mu$, where $l \in \mathbb{R}^n$ is a constant vector

I

2 $D(l^T \mathbf{X}) = l^T \Sigma l$

3 $E(\mathbf{A}\mathbf{X}) = \mathbf{A}\mu$, where $\mathbf{A} \in \mathbb{R}^{p \times n}$ is a constant matrix

4 $D(\mathbf{A}\mathbf{X}) = \mathbf{A}\Sigma\mathbf{A}^T$ and $\text{Cov}(\mathbf{A}\mathbf{X}, \mathbf{B}\mathbf{X}) = \mathbf{A}\Sigma\mathbf{B}^T$

5 If $\text{Cov}(U_p, V_q) = \Gamma$ then $\text{Cov}(\mathbf{A}\mathbf{U}, \mathbf{B}\mathbf{V}) = \mathbf{A}\Gamma\mathbf{B}^T$

implies

implies

implies

$D(x)$ is a p.s.d matrix

Theorem:

Let $\mathbf{X}$ be a random vector with n -components such that $E(x) = \mu$ & $D(x) = \Sigma$. Show that $E(\mathbf{x}^T \mathbf{A} \mathbf{x}) = \text{trace}(\Sigma \mathbf{A}) + \mu^T \mathbf{A} \mu$

(I) Let $\mathbf{A} = \mathbf{I}_n$

$$\begin{aligned} E(\mathbf{x}^T \mathbf{A} \mathbf{x}) &= E(\mathbf{x}^T \mathbf{I}_n \mathbf{x}) \\ &= E(\mathbf{x}^T \mathbf{x}) = E(\sum x_i^2) \\ &= \sum E(x_i^2) \\ &= \sum (\sigma_i^2 + \mu_i^2) \\ &= \sum \sigma_i^2 + \sum \mu_i^2 \\ &\quad \downarrow \quad \quad \downarrow \\ &\quad \text{tr}(\Sigma \mathbf{I}_n) \quad \mu^T \mathbf{I}_n \mu \\ &= \text{tr}(\Sigma \mathbf{I}_n) + \mu^T \mathbf{I}_n \mu \end{aligned}$$

(II) $x_i \stackrel{\text{iid}}{\sim} N(0, 1)$

$\Sigma = \mathbf{I}_n$

$\mathbf{A} = \mathbf{I}_n, \mu_i = 0$

$E(\mathbf{x}^T \mathbf{A} \mathbf{x}) = E(\mathbf{x}^T \mathbf{I}_n \mathbf{x}) = E(\mathbf{x}^T \mathbf{x}) = E(\sum x_i^2)$

Also $\sum_{i=1}^n x_i^2 \sim \chi_n^2$

Chi-Square Distribution

$$= \sum_{i=1}^n (\sigma_i^2 + \mu_i^2)$$

$$= n$$

Say mean $\underline{\mu} \neq 0$

$$E\left(\sum_{i=1}^n x_i^2\right) = E(\sigma_i^2 + \mu_i^2)$$

$$= E(\sigma_i^2) + E(\mu_i^2)$$

$$= (n + \underline{\mu}^T \underline{\mu}) > n$$

Now

$$x_i \sim N(0, 1)$$

$$Q_1 = \underline{x}^T \underline{x} = \sum_{i=1}^n x_i^2$$

Q_1 follows Chi-Square distribution with n df = 0.

$$E(Q_1) = n$$

$$y_i \sim N(\mu_i, 1)$$

$$Q_2 = \underline{y}^T \underline{y} = \sum_{i=1}^n y_i^2$$

$$E(Q_2) = n + \underline{\mu}^T \underline{\mu}$$

non-centrality parameter

Q_2 follows Chi-Square distribution with n df $\neq 0$.

Now we will derive the theorem properly

$$E(\underline{x}^T A \underline{x}) = \text{tr}(\Sigma A) + \underline{\mu}^T A \underline{\mu}$$

Since $\underline{x}^T A \underline{x}$ is a real no. (by simple matrix dimension analysis)

$$\underline{x}^T A \underline{x} = \text{tr}(\underline{x}^T A \underline{x})$$

$$E(\underline{x}^T A \underline{x}) = E(\text{tr}(\underline{x}^T A \underline{x})) = E(\text{tr}(A \underline{x} \underline{x}^T))$$

$$\text{tr}(A E(\underline{x} \underline{x}^T))$$

$$\text{tr}(A(\Sigma + \underline{\mu} \underline{\mu}^T))$$

$$\text{tr}(A \Sigma) + \text{tr}(A \underline{\mu} \underline{\mu}^T)$$

$$= \text{tr}(\Sigma A) + \underline{\mu}^T A \underline{\mu}$$

Theorem:

Let $\underline{x}$ be a random vector with n -components such that $E(\underline{x}) = \underline{\mu}$, $D(\underline{x}) = \Sigma$
 then $P((\underline{x} - \underline{\mu}) \in C(\Sigma)) = 1$

Proof:

NOT CLEAR

Definition:

A random vector X is said to follow multivariate normal $N(\mu, \Sigma)$ if it has a density

$$f(x) = \frac{\exp\left\{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right\}}{(\sqrt{2\pi})^n \sqrt{|\Sigma|}} \quad \begin{array}{l} \mu \in \mathbb{R}^n \\ \text{p.s.d } \Sigma \end{array}$$

Note:- If $X \sim N(\mu, \Sigma)$
 $AX \sim N(A\mu, A\Sigma A^T)$

Definition:

If $X \sim N(\mu, I_n)$ then $X^T X$ is said to follow Chi-square distribution

with degrees of freedom (n) and non-centrality parameter (n.c.p) $\mu^T \mu$.

If $X \sim N(\mu, I_n)$, $E(X^T X) = n + \mu^T \mu$

Definition:

If $X \sim N(\mu, I_n)$ then $X^T A X$ has Chi-Squared distribution iff A is idempotent. Moreover

$$X^T A X \sim \chi^2_{df = \text{Rank}(A), ncp = \mu^T A \mu}$$

Corollary: If A and A_1 are symmetric and idempotent matrices such that $A = A_1 + A_2$ and A_2 is a p.s.d. matrix then $X^T A_1 X$ and $X^T A_2 X$ are independently distributed.

Definition:

Let $X \sim N(\mu, I_n)$ and A is symmetric and $CA = 0$ then $X^T A X$ and CX are independently distributed.

Definition:

Let $X \sim N(\mu, I_n)$ and $X^T A X \equiv \sum_{i=1}^k X^T A_i X$ A_i 's: symmetric, A is idempotent matrix. Then $X^T A_i X \sim \chi^2_{\text{Rank}(A_i), \mu^T A_i \mu}$ and they are independent.

T-statistic

Let $X_i \stackrel{i.i.d}{\sim} N(\mu, \sigma^2)$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

① Aim is to find distribution of $\bar{X}$ & S^2 .

$$X \sim N(\mu, \sigma^2 I_n)$$

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} \underset{\substack{\downarrow \\ C}}{\underset{\sim}{1}^T} \underset{\sim}{X}$$

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = X^T \underset{\substack{\downarrow \\ A}}{\left(I_n - \frac{1}{n} \underset{\sim}{1} \underset{\sim}{1}^T \right)} \underset{\sim}{X}$$

Verify: ① $S^2 = X^T A X$

$$\text{② } A^T = A$$

$$\text{③ } A^2 = A$$

$$\text{④ } \text{Rank}(A) = n-1$$

COCHRAN'S
THEOREM

Now

$$CX \sim N(C\mu, C\Sigma C^T)$$

$$C\mu = \frac{1}{n} \mathbf{1}^T \mu = \frac{1}{n} \mathbf{1}^T (\mu) \mathbf{1} = \mu$$

$$C\Sigma C^T = \left(\frac{1}{n}\right)^2 \mathbf{1}^T \Sigma \mathbf{1} = \frac{1}{n^2} \mathbf{1}^T (\sigma^2 \mathbf{I}_n) \mathbf{1} = \frac{\sigma^2}{n}$$

$$\bar{X} \sim N\left(\mu, \sigma^2/n\right)$$

Now

$$X^T A X = \sigma^2 \left[\left(\frac{X}{\sigma}\right)^T A \left(\frac{X}{\sigma}\right) \right]$$

$$\frac{X}{\sigma} \sim N\left(\frac{\mu}{\sigma}, \mathbf{I}_n\right)$$

Applying
Cochran's
Theorem

$$= \sigma^2 \left(\chi^2_{\text{dof} = n-1, \text{ncp} = \mu^* A \mu = 0} \right)$$

② Show that $\bar{X}$ & s^2 are independently distributed

$$\bar{X} \sim N(\mu, \sigma^2/n), \quad \frac{\bar{X} - \mu}{\sqrt{\sigma^2/n}} \sim N(0, 1)$$

$$\frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{\sqrt{\frac{X^2_{n-1}}{(n-1)\sigma^2}}} \Rightarrow \frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{\sqrt{\frac{X^T A X}{n-1}}} \sim t_{n-1}$$

SIMPLE LINEAR REGRESSION:

$$D = \{(x_i, y_i) \mid x_i \in \mathbb{R}, y_i \in \mathbb{R}, \forall i = 1, 2, \dots, n\}$$

TRUE
ERROR

non-stochastic stochastic

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

→ ϵ_i 's $\equiv$ iid $N(0, \sigma^2)$

unknown parameters: $\beta_0, \beta_1, \sigma^2$

→ we have already found the $\hat{\beta}_0, \hat{\beta}_1$ which minimizes the least squared error function.

→ ESTIMATED ERROR

expression for error $e_i = y_i - \hat{y}_i$

Residual sum
of square

$$\begin{aligned} \sum_i e_i^2 &= \sum (y_i - \hat{y}_i)^2 \\ &= \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \\ &= \sum (y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i)^2 \end{aligned}$$

$$= \sum ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}))^2$$

$$\Rightarrow \sum \left[(y_i - \bar{y})^2 + \hat{\beta}_1^2 (x_i - \bar{x})^2 - 2\hat{\beta}_1 (y_i - \bar{y})(x_i - \bar{x}) \right]$$

$$\Rightarrow S_{yy} + \hat{\beta}_1^2 S_{xx} - 2\hat{\beta}_1 S_{xy}$$

substituting $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$

$$\Rightarrow S_{yy} + \frac{S_{xy}^2}{S_{xx}^2} (S_{xx}) - 2 \frac{S_{xy}}{S_{xx}} S_{xy}$$

$$\Rightarrow S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

We estimate σ^2 by $\hat{\sigma}^2 = \frac{SSR}{n-2} \equiv MSR$ (mean residual sum of square)

Definition:

If an estimator $T(y)$ can be expressed as a linear combination of y with non-random coefficients i.e.

$$T(y) = \sum_i \alpha_i y_i, \text{ then } T(y) \text{ is called a linear estimator.}$$

→ $\hat{\beta}_1$ is a linear estimator of β_1

$$\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{S_{xx}})$$

→ $\hat{\beta}_0$ is a linear estimator of β_0

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

→ Regression line passes through $(\bar{x}, \bar{y})$

→ $SSR \sim \sigma^2 \chi_{n-2}^2$



When regressor is stochastic

SLR

B. Banerjee

SLR

Least square

Estimation

Prediction

Bivariate normal($\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho$)

If (X, Y) has joint density function $f(x, y)$ as

$$f(x, y) = \frac{e^{-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_x}{\sigma_x} \right)^2 + \left(\frac{y-\mu_y}{\sigma_y} \right)^2 - 2\rho \left(\frac{x-\mu_x}{\sigma_x} \right) \left(\frac{y-\mu_y}{\sigma_y} \right) \right]}}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}$$

then (X, Y) is said to follow Bivariate normal.

- If (X, Y) follow Bivariate normal($\mu_x, \mu_y, \sigma_x^2, \sigma_y^2, \rho$) then show that $Y|X = x$ follows $N(\mu_y + \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x), (1 - \rho^2)\sigma_y^2)$
- Perform a large and small sample test for $H_0 : \rho = 0$ Vs $H_1 : \rho \neq 0$.
- Perform a large sample test for $H_0 : \rho = \rho_0$ Vs $H_1 : \rho \neq \rho_0$.

Simple Linear Regression:-

$$\text{Data} = \{ (x_i, y_i) \mid i=1, 2, \dots, n \}$$

$$\text{Model: } y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad \left. \vphantom{y_i} \right\} \text{Model assumption according to SLR.}$$

Assumption:

$$\rightarrow \epsilon_i \stackrel{i.i.d}{\sim} N(0, \sigma^2)$$

$\rightarrow x_i$'s are non-stochastic

Unknown: $\beta_0, \beta_1, \sigma^2$ } have to be estimated.

$$\underset{\sim}{Y} = X \underset{\sim}{\beta} + \underset{\sim}{\epsilon}$$

$(\underset{\sim}{\beta}, \sigma^2)$ to be estimated

Now for parameter estimation we have $\begin{cases} \text{Least Squared Error} \\ \text{Maximum Likelihood Estimation} \end{cases}$

LEAST SQUARED

$$\underset{\sim}{Y} = X \underset{\sim}{\beta} + \underset{\sim}{\epsilon}$$

$$\underset{\sim}{\epsilon} \sim N(0, \sigma^2 I_n) \quad \underset{\sim}{Y} \sim N(X \underset{\sim}{\beta}, \sigma^2 I_n)$$

$$\underset{\sim}{Y} - X \underset{\sim}{\beta} = \underset{\sim}{\epsilon} \sim N(0, \sigma^2 I_n)$$

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

$$= (\underset{\sim}{Y} - X \underset{\sim}{\beta})^T (\underset{\sim}{Y} - X \underset{\sim}{\beta})$$

$$\Rightarrow [\underset{\sim}{Y}^T - (X \underset{\sim}{\beta})^T] [\underset{\sim}{Y} - X \underset{\sim}{\beta}]$$

$$\Rightarrow \underset{\sim}{Y}^T \underset{\sim}{Y} - \underset{\sim}{Y}^T X \underset{\sim}{\beta} - \underset{\sim}{\beta}^T X^T \underset{\sim}{Y} + \underset{\sim}{\beta}^T X^T X \underset{\sim}{\beta}$$

$$\Rightarrow \underset{\sim}{Y}^T \underset{\sim}{Y} - 2 \underset{\sim}{Y}^T X \underset{\sim}{\beta} + \underset{\sim}{\beta}^T (X^T X) \underset{\sim}{\beta}$$

↓ differentiating
w.r.t $\underset{\sim}{\beta}$

$$\frac{\partial S(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta}} = 0$$

$$\frac{\partial (b^T A)}{\partial b} = A^T, \quad \frac{\partial (b^T A b)}{\partial b} = b^T (A + A^T)$$

$$\frac{\partial S(\beta)}{\partial \beta} = 0 \Rightarrow -2y^T x + \beta^T (x^T x + x^T x)$$

$$y^T x = \beta^T x^T x$$

$$x^T y = x^T x \beta$$

$$\beta = (x^T x)^{-1} x^T y$$

Maximum Likelihood Estimate:

$$\begin{aligned} \log(f(\xi)) &= \text{const} \cdot -\frac{1}{2\sigma^2} (y - x\beta)^T (y - x\beta) \\ &= \text{const} \cdot -\frac{1}{2\sigma^2} S(\beta) \end{aligned}$$

to be maximised $\Rightarrow S(\beta)$ to be minimized

under normality condition $\Rightarrow$ errors are normally distributed.

$$\hat{\beta}_{LS} = \hat{\beta}_{MLE}$$

Gauss Markov Model:-

$$y = x\beta + \epsilon$$

$$\epsilon \sim N(0, \sigma^2 I_n)$$

$$y \sim N(x\beta, \sigma^2 I_n)$$

$$\hat{\beta} = (x^T x)^{-1} x^T y$$

Predicted vector $\hat{y}$ for the given old data x is

$$\begin{aligned} \hat{y} = x \hat{\beta} &= [x (x^T x)^{-1} x^T] y \\ &\downarrow \\ P_X y \end{aligned}$$

P_X is the orthogonal projection matrix of column space of X .

$\rightarrow$ The true error i.e. ϵ can not be observed. But we can get the estimated error

$$\hat{\epsilon} = y - \hat{y} = y - P_X y = (I_n - P_X) y$$

→ we know projection matrix is symmetric & idempotent

$$P_X = P_X^T, \quad P_X = P_X^2$$

$$\rightarrow P_X (I_n - P_X) = 0$$

$$\begin{aligned} \rightarrow e^T \hat{y} &= [(I_n - P_X) Y]^T P_X Y \\ &= Y^T (I_n - P_X) P_X Y \\ &\Rightarrow Y^T \underbrace{(P_X - P_X^2)}_0 Y = 0 \quad [\text{error is normal to } \hat{y}] \end{aligned}$$

→ let's find distribution of $\hat{\beta}$.

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad Y \sim N(X\beta, I_n \sigma^2)$$

let's find distribution of $\hat{\beta}$

$$(X^T X)^{-1} X^T Y \sim N \left((X^T X)^{-1} X^T (X\beta), (X^T X)^{-1} X^T I_n \sigma^2 X (X^T X)^{-1} \right)$$

$$\hat{\beta} \sim N \left(\beta, (X^T X)^{-1} X^T I_n X (X^T X)^{-1} \sigma^2 \right)$$

$$\hat{\beta} \sim N \left(\beta, \cancel{(X^T X)^{-1}} \cancel{(X^T X)} (X^T X)^{-1} \sigma^2 \right)$$

$$\hat{\beta} \sim N \left(\beta, (X^T X)^{-1} \sigma^2 \right)$$

→ finding distribution of e

$$e = (I_n - P_X) Y$$

$$Y \sim N(X\beta, I_n \sigma^2)$$

$$e \sim N \left((I_n - P_X) X\beta, (I_n - P_X) I_n \sigma^2 (I_n - P_X)^T \right)$$

$$e \sim N \left(\underbrace{(I - P_X) X\beta}_0, (I_n - P_X) \sigma^2 \right)$$

$$\sim N(0, (I_n - P_X) \sigma^2)$$

→ Predicted & Estimated Error

$$\hat{y} = X \hat{\beta} = P_X Y \rightarrow \text{predicted}$$

$$\hat{e} = (I_n - P_X) \underline{y} \rightarrow \text{estimated error}$$

$$\text{Cov}(\hat{y}, \hat{e}) = \text{Cov}(P_X \underline{y}, (I_n - P_X) \underline{y})$$

$$\text{Cov}(A\underline{x}, B\underline{x}) = A \Sigma B^T$$

$$\therefore \text{Cov}(P_X \underline{y}, (I_n - P_X) \underline{y}) = P_X (I_n \sigma^2) (I_n - P_X)^T$$

$\downarrow$
 $\Sigma \text{ for } \underline{y} \quad (\because \underline{y} \sim N(X\beta, I_n \sigma^2))$

$$= 0$$

Simple Linear Regression

$$T(\underline{y}) = \sum_{i=1}^n \alpha_i y_i = \underline{\alpha}^T \underline{y}$$

$\downarrow$
 linear estimator

We can show that $\hat{\beta}_1$ is a linear estimator:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum y_i (x_i - \bar{x})}{S_{xx}}$$

$$\hat{\beta}_1 = \sum \underbrace{\left(\frac{x_i - \bar{x}}{S_{xx}} \right)}_{\alpha_i} y_i = \underline{\alpha}^T \underline{y}$$

We have already found the distribution of $\hat{\beta}_1$, let's try again with a different approach:

$$\underline{y} \sim N(X\beta, I_n \sigma^2)$$

$$\hat{\beta}_1 \sim \underline{\alpha}^T \underline{y}$$

$$\hat{\beta}_1 \sim N(\underline{\alpha}^T X \beta, \underline{\alpha}^T (I_n \sigma^2) \underline{\alpha})$$

mean

$$\underline{\alpha}^T \begin{bmatrix} 1 & \bar{x} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$$\begin{bmatrix} \cancel{\frac{\sum (x_i - \bar{x})}{S_{xx}}} & \cancel{\frac{\sum (x_i - \bar{x}) x_i}{S_{xx}}} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$\downarrow$
 0

$$\sum \frac{(x_i - \bar{x}) x_i}{S_{xx}} = 1$$

$$\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \beta_1$$

variance

$$\underline{\alpha}^T (I_n \sigma^2) \underline{\alpha} = \sigma^2 (\underline{\alpha}^T \underline{\alpha})$$

$$= \sigma^2 \cdot \frac{\sum (x_i - \bar{x})^2}{S_{xx}^2}$$

$$= \sigma^2 \cdot \frac{S_{xx}}{S_{xx}^2}$$

$$= \frac{\sigma^2}{S_{xx}}$$

β_1

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

Homework

(a)

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_0 = \bar{y} - \frac{\sum y_i (x_i - \bar{x})}{S_{xx}} \bar{x}$$

$$= \frac{\sum y_i}{n} - \frac{\sum y_i (x_i - \bar{x})}{S_{xx}} \bar{x}$$

$$\Rightarrow \sum \left(\frac{1}{n} - \frac{\bar{x} (x_i - \bar{x})}{S_{xx}} \right) y_i$$

$\downarrow$
 α_i

(b)

$$\hat{\beta}_0 \sim \alpha^T y$$

$$\hat{\beta}_0 \sim N\left(\alpha^T X \beta, \alpha^T I_n \sigma^2 \alpha\right)$$

variance

mean

$$\alpha^T \begin{bmatrix} 1 \\ \bar{x} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$$\left[\sum \left(\frac{1}{n} - \frac{\bar{x} (x_i - \bar{x})}{S_{xx}} \right), \sum \left[\frac{x_i}{n} - \frac{\bar{x} (x_i - \bar{x})}{S_{xx}} \right] \right]$$

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \beta_0$$

$$\alpha^T I_n \sigma^2 \alpha$$

$$\sigma^2 \alpha^T \alpha$$

$$\sigma^2 \sum \left(\frac{1}{n} - \frac{\bar{x} (x_i - \bar{x})}{S_{xx}} \right)^2$$

$$\sigma^2 \sum \left[\frac{1}{n^2} + \frac{\bar{x}^2 (x_i - \bar{x})^2}{S_{xx}^2} - \frac{2\bar{x} (x_i - \bar{x})}{n S_{xx}} \right]$$

$$\sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2 S_{xx}}{S_{xx}^2} \right]$$

$$\sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right]$$

$$\hat{\beta}_0 \sim N\left[\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)\right]$$

(*) Estimation of σ^2

$$e \sim N(0, \sigma^2 I_n) \quad \text{not observable}$$

$$e = y - \hat{y} = (I_n - P_X) y \sim N\left(0, (I_n - P_X) \sigma^2\right)$$

ε_i are not independently distributed.

$$\text{let } \hat{\sigma}^2 = \frac{e^T e}{n-2}$$

$$E(\hat{\sigma}^2) = \sigma^2$$

$\hat{\sigma}^2$ is an unbiased estimate of σ^2 .

$$\text{we already know:- } \sum e^2 = S_{yy} - \frac{S_{xy}^2}{S_{xx}}$$

MULTIPLE LINEAR REGRESSION

$$Y \sim N(X\beta, \sigma^2 I_n)$$

say we have k features $\Rightarrow (\beta_0, \beta_1, \dots, \beta_k)$ $\xrightarrow{\sigma^2}$ $k+2$ parameters

LUE: A linear estimator $L^T y = \sum l_i y_i$ is said to be linear unbiased estimator (LUE) of $P^T \beta$ for $\forall \beta \in \mathbb{R}^{k+1}$.

LZF: A linear estimator $L^T y = \sum l_i y_i$ is said to linear zero function (LZF) if $E(L^T y) = 0$, $\forall \beta \in \mathbb{R}^{k+1}$

BLUE: The Best Linear Unbiased Estimator (BLUE) of a linear parameter function $P^T \beta$ is a LUE with minimum variance.

$\hookrightarrow$ A linear function is BLUE of it's expectation iff it is uncorrelated with all LZF.

$\hookrightarrow$ If $L^T y$ is LUE of $P^T \beta$, then $L^T P x y$ is the BLUE of $P^T \beta$.

$\hookrightarrow$ The if and only if condition for $P^T \beta$ to be estimable is $P \in C(X^T)$

$\hookrightarrow L^T y$ is a LZF if and only if $X^T l = 0$

$\hookrightarrow$ BLUE is Unique.

ANOVA

⑤

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$H_1 : H_0$ is not true

Under H_0 $y_i \sim N(\beta_0, \sigma^2)$

MATRIX FORM

$$\rightarrow SS_{Total} = \sum (y_i - \bar{y})^2 = \underline{y^T (I_n - \frac{1}{n} \mathbf{1} \mathbf{1}^T) y}$$

$$\rightarrow SS_{Model} = \sum (\hat{y}_i - \bar{y})^2 = \underline{y^T (P_X - \frac{1}{n} \mathbf{1} \mathbf{1}^T) y}$$

$$\rightarrow SS_{Error} = \sum (y_i - \hat{y}_i)^2 = \underline{y^T (I_n - P_X) y}$$

$$SS_{Total} = SS_{Model} + SS_{Error}$$

↳ can get by adding & subtracting P_X in SS_{Total} .

DISTRIBUTION

$$\frac{SS_{Error}}{\sigma^2} \sim \chi^2_{\substack{\text{dof} = n-k-1 \\ \text{nep} = 0}}$$

$$\frac{SS_{Model}}{\sigma^2} \sim \chi^2_{\substack{\text{dof} = k \\ \text{nep} = \lambda > 0}}$$

$$\frac{SS_{Total}}{\sigma^2} \sim \chi^2_{\substack{\text{dof} = n-1 \\ \text{nep} = \lambda > 0}}$$

under $H_0 : \lambda = 0$

$H_1 : \lambda > 0$

$$\frac{\left(\frac{SS_{Model}}{\sigma^2} \right) / k}{\left(\frac{SS_{Error}}{\sigma^2} \right) / (n-k-1)} \sim F_{\text{dist'n.}}$$

If $|F_{obs}| > F_{\alpha, k, n-k-1} \Rightarrow \underline{\underline{\text{reject } H_0.}}$

NOTE

II

$$H_0: \beta_j = b_j$$

$$H_1: \beta_j \neq b_j$$

$$\frac{\hat{\beta}_j - b_j}{\sqrt{\hat{\sigma}^2 c_{jj}}} \sim t_{n-k-1}$$

$$t_{n-k-1} > T_{\alpha/2, n-k-1} \text{ reject } H_0.$$

III

$$H_0: \beta_i - \theta \beta_j = 0$$

$$H_1: \beta_i - \theta \beta_j > 0$$

$$T = \frac{\mathbf{l}^T \hat{\boldsymbol{\beta}} - b}{\sqrt{\hat{\sigma}^2 (c_{ii} - 2\theta c_{ij} + \theta^2 c_{jj})}} \sim t_{n-k-1}$$

$$t_{n-k-1} > T_{\alpha, n-k-1}$$

Coefficient of Determination

$$R^2 = \frac{SS_{\text{Model}}}{SS_{\text{Total}}} = 1 - \frac{SS_{\text{Error}}}{SS_{\text{Total}}} = 1 - \frac{\sum e_i^2}{\sum (y_i - \bar{y})^2}$$

Adjusted R^2 :

$$R^2_{\text{adj}} = 1 - \frac{\sum e_i^2 / (n-k-1)}{\sum (y_i - \bar{y})^2 / (n-1)} < R^2$$



does not depend on number of variables.

PROBABILITY & STATISTICS PRE-REQUISITE

→ Standard Normal Distribution - $[X \sim N(\mu, \sigma^2)]$

$$\text{PDF: } \phi(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$E(x) = \mu$$

$$\text{Var}(x) = \sigma^2$$

→ Gamma $[X \sim \text{Gamma}(\alpha, \lambda)]$

$$\text{PDF: } \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$$

$$E(x) = \alpha/\lambda$$

$$\text{Var}(x) = \alpha/\lambda^2$$

$$\begin{aligned} \Gamma(n) &= (n-1)! = \int_0^\infty x^{n-1} e^{-x} dx & \text{erf}(z) &= \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt \\ \Gamma(n+1) &= n \Gamma(n) \\ \Gamma(1/2) &= \sqrt{\pi} \end{aligned}$$

Estimators & Statistics

Population: Is a collection of measurements

Sample: subset of population

Random Sampling: Let $x_1, x_2, \dots, x_n$ be i.i.d. r.v.'s each having same pdf $f(x)$. Then we say that $(x_1, x_2, \dots, x_n)$ is a random sample from population with distribution $f(x)$.

Statistic: A function of a random sample

$T = T(x_1, x_2, \dots, x_n)$ is called a statistic.

Ex:- $\bar{X} = \frac{1}{n} \sum x_i$ (Sample mean)

$$S^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2 \quad (\text{Sample variance})$$

The probability distribution of a statistic is called Sampling Distribution.

Chi-Square Distribution (χ^2):

A continuous r.v X is said to have a Chi-Square distribution with n degrees of freedom if it has pdf given by

$$f_X(x) = \frac{1}{2^{n/2} \Gamma(n/2)} e^{-x/2} \cdot x^{n/2-1}, \quad x > 0, n > 0$$

(special case of Gamma Distribution)

$$G(n/2, 1/2)$$

$$E(X) = n/2 / 1/2 = n$$

$$\text{Var}(X) = n/2 / (1/2)^2 = 2n$$

→

$$X \sim N(0, 1)$$

$$Y = X^2 \Rightarrow Y \sim \chi_1^2$$

→ let $X_1, X_2, \dots, X_n$ be i.i.d $N(0, 1)$

$$\text{Then } Y = \sum_{i=1}^n X_i^2 \sim \chi_n^2$$

If we are doing sampling from a standard normal distribution then the distribution of the sample variance is a chi-square distribution.

T-distribution:

let X and Y be two independent random variables.

$$X \sim N(0, 1), Y \sim \chi_n^2. \text{ Then}$$

$$T = \frac{X}{\sqrt{\frac{Y}{n}}} = \frac{\sqrt{n} X}{\sqrt{Y}} \text{ is said to have a student's } t\text{-distribution on } n\text{-degrees of freedom.}$$

$$\frac{\sqrt{n} (\bar{X} - \mu)}{S} \sim t_{n-1}$$

F-distribution

let Y_1 and Y_2 be independently distributed & $Y_1 \sim \chi_m^2$

$$Y_2 \sim \chi_n^2$$

Then $U = \frac{Y_1/m}{Y_2/n} = \frac{n}{m} \frac{Y_1}{Y_2}$ is said to have an F-dist'n on

(m, n) degrees of freedom.

Let $X_1, X_2, \dots, X_m \overset{\text{i.i.d}}{\sim} N(\mu_1, \sigma_1^2)$
 $Y_1, Y_2, \dots, Y_n \overset{\text{i.i.d}}{\sim} N(\mu_2, \sigma_2^2)$
indep'n

$$S_x^2 = \frac{1}{m-1} \sum (X_i - \bar{X})^2$$

$$S_y^2 = \frac{1}{n-1} \sum (Y_i - \bar{Y})^2$$

$$U = \frac{(m-1)S_x^2}{\sigma_1^2} \sim \chi_{m-1}^2, \quad V = \frac{(n-1)S_y^2}{\sigma_2^2} \sim \chi_{n-1}^2$$

$$\frac{V/(n-1)}{V/(n-1)} = \frac{\sigma_1^2}{\sigma_2^2} \frac{S_x^2}{S_y^2} \sim F_{m-1, n-1}$$

If $T \sim t_n$. Then $T^2 \sim F_{1, n}$

Problems of Statistical Inference:

- ① Point Estimation
- ② Interval Estimation
- ③ Testing of Hypothesis

Basic Terminology:- Population, Sample, Parameter, Statistic, Estimator, Estimate.

Point Estimation

$x_1, x_2, \dots, x_n$ is a random sample from a population with distribution say P_θ , $\theta \in \Theta$.

P means we know the distribution & we focusing on parametric estimation.

θ is the parameter, Θ = parameter space

Ex:- ① $x_1, x_2, \dots, x_n \stackrel{i.i.d}{\sim} N(\mu, \sigma^2)$, $\theta = (\mu, \sigma^2)$
 $\Theta = \mathbb{R} \times \mathbb{R}^+$

② $x_1, x_2, \dots, x_n \sim P(\lambda)$ $\lambda > 0$
 $\Theta = \mathbb{R}^+$

$$T(x_1, x_2, \dots, x_n) = T(\underline{x})$$

Expression for a statistic
 $\downarrow$
 to be used as an estimator.

Unbiased Estimation

A statistic $T(\underline{x})$ is said to be an unbiased estimator of $g(\theta)$ if

$$E_\theta T(\underline{x}) = g(\theta) \quad \forall \theta \in \Theta$$

$g(\theta)$: some parametric function.

If $E_\theta T(\underline{x}) = g(\theta) + b(\theta)$ then we say that $T(\underline{x})$ is biased for $g(\theta)$ and $b(\theta)$ is the bias of $T(\underline{x})$.

Example 1: $X \sim \text{Bin}(n, p)$ n is known
 $0 \leq p \leq 1$

$$\text{Let } T(x) = \frac{x}{n}, \quad E\left(\frac{x}{n}\right) = \frac{1}{n} E(x) = \frac{1}{n} (np) = p$$

So $\frac{x}{n}$ is unbiased for p .

$$\begin{aligned}
 \textcircled{\#} \quad E(X(X-1)) &= E(X^2 - X) = E(X^2) - E(X) \\
 &= np(1-p) + (np)^2 - np \\
 &= \cancel{np} - np^2 + n^2p^2 - \cancel{np} \\
 &= np^2(n-1) \\
 &= n(n-1)p^2
 \end{aligned}$$

$$E\left(\frac{X(X-1)}{n(n-1)}\right) = p^2$$

estimator

$$\begin{aligned}
 \text{Var}(X) &= np(1-p) \\
 &= n(p - p^2) \\
 &= n\left(\frac{X}{n} - \frac{X(X-1)}{n(n-1)}\right) = E\left\{\frac{X(n-X)}{n-1}\right\} = np(1-p)
 \end{aligned}$$

So $\frac{X(n-X)}{n-1}$ is an unbiased estimator for $\text{Var}(X)$.

Consistency: An estimator $T_n = T(\underline{X})$ is said to be consistent for $g(\underline{\theta})$ if for every $\epsilon > 0$

$$P(|T_n - g(\underline{\theta})| > \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

Least Square Estimation

$$(x_1, y_1) \quad (x_2, y_2) \quad \dots \quad (x_n, y_n)$$

$$y_i = \alpha + \beta x_i + \epsilon_i \quad i=1, 2, \dots, n$$

sum of squares error

$$S = \sum \epsilon_i^2 = \sum [y_i - (\alpha + \beta x_i)]^2$$

$$\hat{\beta} = \frac{S_{yx}}{S_{xx}}, \quad \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$$

These are least squared estimates of α & β .

If we assume $\epsilon_i \stackrel{i.i.d}{\sim} N(0, \sigma^2)$

$$\text{then } y_i \sim N(\alpha + \beta x_i, \sigma^2)$$

$$E(\hat{\beta}) = E\left(\frac{S_{yx}}{S_{xx}}\right) = \frac{1}{S_{xx}} E(S_{yx}) = \beta \frac{\cancel{S_{xx}}}{\cancel{S_{xx}}} = \beta$$

$$S_{yx} = \sum x_i y_i - n \bar{x} \bar{y}$$

$$E(S_{yx}) = \sum x_i E(y_i) - n \bar{x} E(\bar{y})$$

$$= \sum x_i (\alpha + \beta x_i) - n \bar{x} (\alpha + \beta \bar{x})$$

$$= \cancel{n \bar{x} \alpha} + \beta \sum x_i^2 - \cancel{n \bar{x} \alpha} - n \beta \bar{x}^2$$

$$= \beta \left(\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right)$$

$$= \beta \left(\sum x_i^2 - \frac{(\sum x_i)^2}{n} \right)$$

$$= \beta S_{xx}$$

$\hat{\beta}$ is an unbiased estimator of β .

$$E(\hat{\alpha}) = E(\bar{y} - \hat{\beta}\bar{x}) = \alpha + \cancel{\beta\bar{x}} - \cancel{\hat{\beta}\bar{x}} = \alpha$$

$\hat{\alpha}$ is an unbiased estimator of α .

So the least square estimators of α and β are unbiased for α & β respectively.

$$E\left(\frac{SSE}{n-2}\right) = \sigma^2$$

$MSE = \frac{SSE}{n-2}$ is an unbiased estimator of σ^2 .

Method of Maximum Likelihood Estimation

When the sample $(x_1, x_2, \dots, x_n)$ is observed corresponding to r.v.'s $X_1, \dots, X_n$ then the joint pdf or pmf

$f(x_1, x_2, \dots, \underline{\theta})$ is called the likelihood for $L(\underline{\theta}, \underline{x})$

$$\underline{x} = (x_1, \dots, x_n).$$

Then the maximum likelihood estimator for $\underline{\theta}$ is that value $\hat{\underline{\theta}}(\underline{x})$ for which

$$L(\hat{\underline{\theta}}(\underline{x}), \underline{x}) \geq L(\underline{\theta}, \underline{x}) \quad \forall \underline{\theta} \in (H)$$